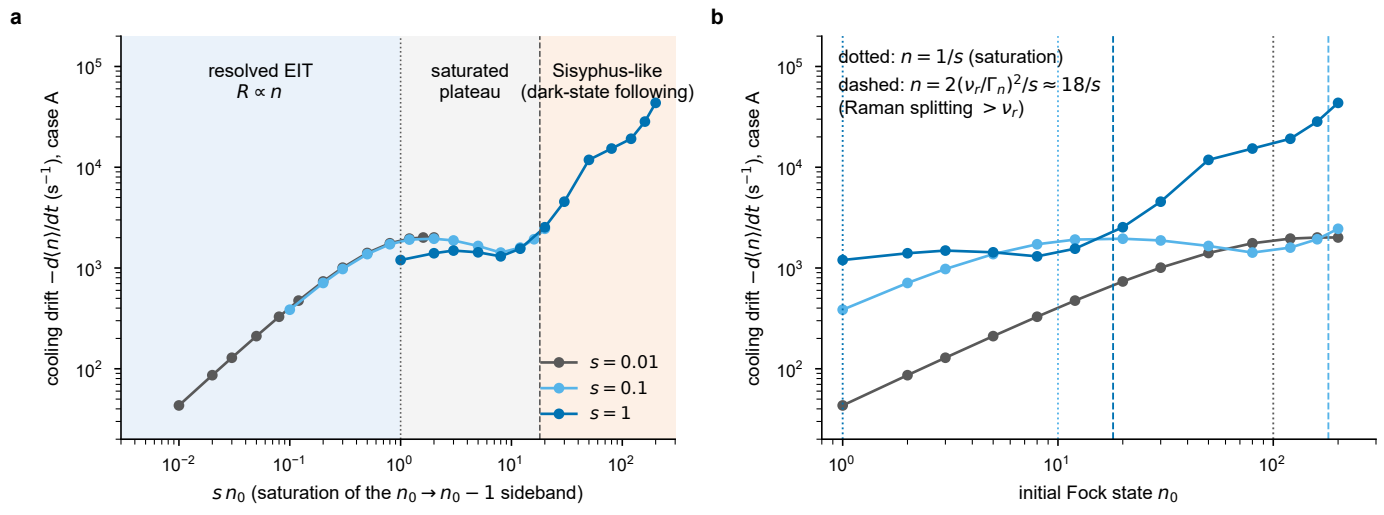

a
resolved EIT
$R \propto n$
saturated
plateau
Sisyphus-like
(dark-state following)
$s = 0.01$
$s = 0.1$
$s = 1$
cooling drift $-d\langle n\rangle/dt$ (s^{-1}), case A
$s\,n_0$ (saturation of the $n_0 \to n_0 - 1$ sideband)
b
dotted: $n = 1/s$ (saturation)
dashed: $n = 2(\nu_r/\Gamma_n)^2/s \approx 18/s$
(Raman splitting $> \nu_r$)
cooling drift $-d\langle n\rangle/dt$ (s^{-1}), case A
initial Fock state n_0